Worksheet 7

Integrated cases – DETAILED SOLUTIONS

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Detailed solution – Integrated case A

Observed population: 200 students; unit: one student; variable: maximum monthly acceptable price; type: continuous quantitative.

Class	$[5, 10[$	$[10, 15[$	$[15, 20[$	$[20, 30[$	$[30, 40[$	$[40, 60[$
n_i	10	25	45	60	40	20
Width	5	5	5	10	10	20
Density	2	5	9	6	4	1
Relative frequency (%)	5	12.5	22.5	30	20	10
ICC	10	35	80	140	180	200
DCC	200	190	165	120	60	20
Midpoint	7.5	12.5	17.5	25	35	50

Modal class: $[15, 20[$. Median:

$$20 + 10(100 - 80)/60 = \boxed{23.33}.$$

$$Q_1 = 15 + 5(50 - 35)/45 = \boxed{16.67}, \quad Q_3 = 30 + 10(150 - 140)/40 = \boxed{32.50}.$$

$$\bar{x} = 5075/200 = \boxed{25.375}.$$

With $\sum n_i c_i^2 = 154750$,

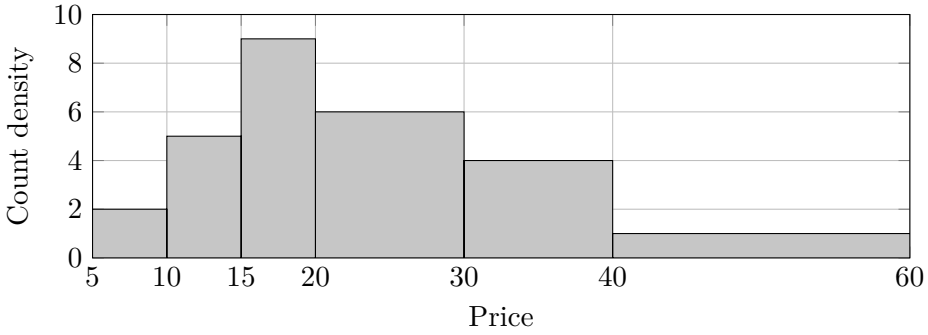
$$V = 154750/200 - 25.375^2 = \boxed{129.8594},$$

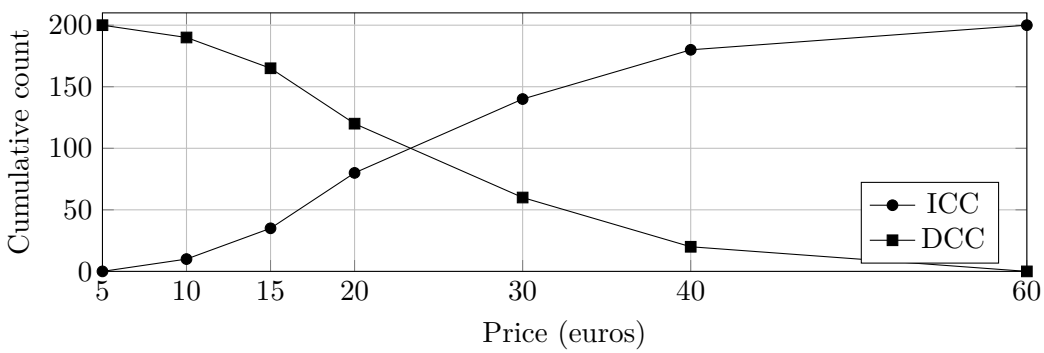
$$\sigma = \boxed{11.396}, \quad CV = \boxed{44.91\%}.$$

For 65% to afford the price, 35% must be below it: target count 70. Interpolation in $[15, 20[$ gives

$$15 + 5(70 - 35)/45 = \boxed{18.89 \text{ euros}}.$$

Satisfaction is qualitative; bar/pie charts are suitable; mean and standard deviation are not appropriate.





Detailed solution – Integrated case B, Part 1

$$\sum x = 73,\ \sum y = 146,\ \sum x^2 = 769,\ \sum y^2 = 3106,\ \sum xy = 1545.$$

This gives $R = \boxed{0.9983}$ and

$$\boxed{\hat{y} = 2.06804x - 0.62090}.$$

For $x = 18$, $\hat{y} = 36.6039$, i.e. about $\boxed{366\,039 euros}$. **Detailed solution – Integrated case B, Part 2**

$$\sum p_0q_0 = 3200,\quad \sum p_0q_1 = 3410,\quad \sum p_1q_0 = 3620,\quad \sum p_1q_1 = 3861.$$

$$F^p = \boxed{113.175},\quad F^q = \boxed{106.610},\quad I_v = \boxed{120.656}.$$

(Intermediate results: $L^p = 113.125$, $P^p = 113.226$, $L^q = 106.563$, $P^q = 106.657$.) **Detailed solution – Integrated case B, Part 3**

$0.80 + 1.10 + 0.90 + 1.20 = 4$, so the average is 1.

t	Trend	Seasonal coefficient	Forecast
13	86	0.80	68.8
14	88	1.10	96.8
15	90	0.90	81.0
16	92	1.20	110.4